Software Requirements Specification

for

QTC Workflow Queue (WFQ)

Version 1.0

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Revision History

Name	Date	Reason For Changes	Version
All group members	10/17/2025	Initial Document	1.0
Slok Patel	10/17/2025	Section 1 and 2 introduction	1.1
Zhen Zhao	10/22/2025	Section 4 Requirement Specifications	1.2
Edwin Rojas	10/23/2025	Section 3 External Interface Requirements	1.2
Slok Patel	10/23/2025	Section 4, 5, & Appendix (ABC)	1.2
Dylan Huang	10/23/2025	Section 6 Legal and Ethical Considerations	1.2
Laila Velasquez	10/23/2025	Table of Contents	1.2
Edwin Rojas	11/4/2025	Section 1, 2, and 3 Revisions	1.3

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the functional and non-functional requirements for the Workflow Queue (WFQ) Administrator Application. This document specifies what the software will do and the conditions under which it will operate. It serves as a guide for developers, testers, and stakeholders by clearly outlining the system's intended features and behaviors. The WFQ application will be developed using ASP.NET MVC with C#, Entity Framework, and stored procedures, with both SQL Server and Oracle serving as supported backends. This SRS focuses on Version 1.0 of the new Administrator Application, which will be developed using ASP.NET MVC with C# and Entity Framework. Its sole backend interaction will be with the existing QTC-provided SQL Server database. This document provides the foundation for the design and implementation activities described in the Software Design Document (SDD).

1.2 Intended Audience and Reading Suggestions

This document is intended for a variety of audiences involved in the project lifecycle:

- **Project Sponsor (Leidos QTC)**: Can use this document to verify that the planned application meets their specified needs and to understand the project's scope and boundaries.

- **Developers**: Should focus on the detailed functional and non-functional requirements in Sections 4 and 5 to guide implementation.

- **QA Testers**: Should use the requirements in Section 4 as a basis for writing test cases and defining acceptance criteria.

- **Project Managers**: Can use the overall product description and defined scope to manage project execution and stakeholder expectations.

1.3 Product Scope

The WFQ Administrator Application is a new, web-based tool designed for authorized technical administrators to create, configure, and manage the workflows used by the main QTC WFQ system. The application will achieve this by directly reading from and writing to the existing WFQ SQL Server database.

The core purpose of this application is to provide a user-friendly graphical interface for managing the underlying data that defines the workflow engine's behavior. The initial release will focus on providing full administrative control over workflows and their associated steps.

To ensure clarity, the project scope is explicitly defined as follows:

IN SCOPE:

- Designing, building, and delivering a new, standalone Administrator web application.

- Creating a data access layer within the new application that communicates directly with the existing WFQ SQL Server database.

- Implementing all administrative features as specified in the "Workflow Queue Administrator Requirements" document.
- Implementing role-based access control within the new application based on a URL query string parameter.

OUT OF SCOPE:
- Accessing, modifying, or using any source code from the existing live WFQ application.
- Calling, updating, or interacting with any pre-existing backend REST API.
- Building or modifying the existing WFQ User & Manager interface.
- Integrating with external third-party services or creating advanced analytics dashboards.

1.4 Definitions, Acronyms, and Abbreviations

- **WFQ** – Workflow Queue
- **ASP.NET MVC** – A Model-View-Controller framework for web applications using Microsoft's .NET platform
- **C#** – Programming language used for the system's logic and business layer
- **SQL Server / Oracle** – Database management systems used for backend storage
- **Entity Framework (EF) –** An object-relational mapping (ORM) framework used for data operations
- **QA** – Quality Assurance
- **PO** – Product Owner
- **SRS** – Software Requirements Specification
- **SDD** – Software Design Document

1.5 References

- Microsoft. "C# Coding Conventions." Retrieved from: https://learn.microsoft.com/en-us/dotnet/csharp/fundamentals/coding-style/coding-conventions
- Microsoft Learn. "Build Web APIs with ASP.NET Core." Retrieved from: https://learn.microsoft.com/en-us/training/modules/build-web-api-aspnet-core/
- Microsoft Learn. "Install SQL Server Management Studio (SSMS)." Retrieved from: https://learn.microsoft.com/en-us/ssms/install/install

- Oracle. “SQL Developer Download.” Retrieved from: https://www.oracle.com/database/sqldeveloper/technologies/download/

2. Overall Description

The WFQ (Workflow Queue) application provides a centralized platform for automating and managing multi-step workflows across organizational departments. Its purpose is to replace inefficient manual processes and disconnected tools with a cohesive system that improves coordination, reduces delays, and enhances visibility. This section gives a high-level overview of the system's purpose, context, and environment, preparing the reader for the detailed requirements described later in the document. The WFQ system focuses on streamlining workflow approvals, ensuring accountability, and supporting data consistency across multiple database environments.

2.1 System Analysis

The WFQ system is designed to solve the problem of inefficient and fragmented workflow management. Many organizations currently use manual methods or isolated applications for approvals, which often result in miscommunication, data loss, and process delays. The WFQ application automates these processes by enabling users to create, track, and complete workflows through a unified interface.

The main goals of the system include increasing process transparency, reducing approval times, and ensuring accountability at every step. Key technical challenges involve maintaining data integrity between SQL Server and Oracle databases and implementing secure, role-based authentication. These challenges will be addressed through the use of Entity Framework, parameterized stored procedures, and industry-standard authentication mechanisms to ensure both performance and security.

2.2 Product Perspective

The WFQ Administrator Application is a new, self-contained, internal web tool. It is not an enhancement to the existing WFQ application but rather a separate, complementary system that shares a common database.

Its position within the QTC ecosystem is that of a "back-office" or "control panel" tool. While the primary WFQ application is used by front-line staff to execute work, this new Administrator Application will be used by technical staff to define and manage the structure of that work. Its modular nature ensures that its development and deployment will not impact the operation of the existing live system.

2.3 Product Functions

The primary functions of the WFQ Administrator Application are focused exclusively on workflow management. These include:

- **Workflow Creation and Configuration**: Allows authorized administrators to define new workflows.
- **Workflow Step Management:** Allows administrators to add, edit, and delete the individual steps that constitute a workflow, including their sequence and properties.
- **Task Oversight**: Provides administrators with the ability to view tasks within a workflow and perform limited management functions, such as reassigning or completing a task.
- **History and Auditing**: Displays a history of tasks and changes within a workflow to support troubleshooting and oversight.

2.4 User Classes and Characteristics

The WFQ system will be used by multiple user groups with different roles and privileges:

- **Administrators:** Manage user accounts, permissions, and system configurations. They are responsible for defining workflow templates and monitoring overall system health.
- **Managers / Approvers:** Review and approve workflow items. They have access to reporting tools and can delegate or escalate tasks when necessary.
- **Regular Users / Submitters:** Initiate workflows, upload relevant documents, and monitor the progress of their requests.
- **QA Testers:** Test application features and ensure compliance with functionality, performance, and security standards.
- **Developers:** Maintain and enhance the system, ensuring compatibility across environments.

Each user class has distinct responsibilities and technical expertise. Administrators and developers are typically more technical, while end users focus on usability and functionality.

2.5 Operating Environment

The WFQ application will operate in a **web-based environment** using the following components:

- **Frontend:** Developed with ASP.NET MVC for the UI layer, compatible with modern browsers (Chrome, Edge, Firefox).
- **Backend:** Built in **C#** with **Entity Framework**, handling data access logic.
- **Databases:** Microsoft **SQL Server** and **Oracle Database** for backend storage and stored procedure execution.
- **Operating System:** Hosted on **Windows Server** with support for Internet Information Services (IIS).
- **Development Tools:** Visual Studio, SQL Server Management Studio (SSMS), and Oracle SQL Developer.

2.6 Design and Implementation Constraints

The development of this application must adhere to the following constraints:

- **Direct Database Access Only:** The application must not attempt to interface with the existing WFQ application at the code or API level. All interactions must occur via direct queries and commands to the SQL Server database**.**
- **Technology Stack:** The project is constrained to the existing technology stack (C#, ASP.NET, MS SQL Server) to ensure maintainability and consistency. No other database systems (e.g., Oracle) or backend languages are in scope for this version.

- **Security & Compliance (HIPAA):** The application will be handling data that is subject to HIPAA regulations. All access must be strictly controlled, and all administrative actions that modify data must be auditable. The design must prioritize data security and confidentiality.

- **Authorization Model:** The application must rely solely on a URL query string parameter for determining user authorization. It will not integrate with a separate authentication service API.

2.7 User Documentation

Upon completion, the WFQ application will include several user documentation resources to assist in deployment and usage:

- **User Manual:** Provides step-by-step guidance for creating, managing, and approving workflows.

- **Online Help Guides:** Integrated help documentation within the application's UI.

- **Developer Documentation:** Includes API references, database schema details, and setup instructions for developers and administrators.

- **Training Materials:** Optional tutorial videos or PDF guides to assist new users in understanding workflow features.

2.8 Assumptions and Dependencies

The project relies on the following assumptions and dependencies:

- **Stable Database Schema**: We assume the schema of the existing WFQ database is stable and will not undergo significant changes during the project lifecycle

- **Database Accessibility**: We assume that the development team will be provided with network access and credentials to a development/testing instance of the WFQ database.

- **Sponsor-Provided Authorization**: We are dependent on the hosting environment to provide a reliable and secure mechanism for passing the user's administrative status via the URL query string.

- **Team Skillset**: We assume the team possesses, or can acquire through independent learning, the necessary skills in C#, ASP.NET MVC, and SQL to complete the project.

2.9 Apportioning of Requirements

Certain advanced features may be reserved for future versions of the WFQ application, depending on project scope and resource availability. These may include:

- Integration with third-party APIs for document signing or notifications (e.g., Outlook, Teams).

- Advanced analytics dashboards using Power BI.

- Mobile version of the WFQ system for iOS and Android platforms.

- Support for multilingual UI and accessibility enhancements.

3. External Interface Requirements

This section defines the interfaces between the new QTC Workflow Queue (WFQ) Administrator Application and the outside world. Based on the clarified project scope, the system has a limited number of external interfaces, primarily with its single user class and the existing database.

3.1 User Interfaces

The system will present a single, new web-based user interface.

1. **Administrator UI:**
    - **Description**: This is a new, dedicated web application built from scratch for technical administrators to manage the workflow system. It will provide all functionality specified in the "Workflow Queue Administrator Requirements" document, including the ability to create, edit, and view workflows and their steps.
    - **Interface Type**: Graphical User Interface (GUI) delivered via a modern web browser (e.g., Chrome, Edge).

GUI and Style Standards:

- **Branding**: All UIs, particularly the new Administrator UI, must adhere to the Leidos corporate branding guidelines, including the official color palette and logos found on the leidos.com website.
- **Layout**: Screen layouts will be standardized. Standard functions (e.g., search, create, edit) will be presented consistently.
- **Error Display**: The system will display user-friendly error messages for invalid inputs or exceptions (e.g., concurrency, foreign key violations) .
- **Accessibility**: The UI design should consider conformance with Section 508 compliance rules.

3.2 Hardware Interfaces

The WFQ Administrator Application is a software-only system. It does not interface directly with any custom or specialized hardware. All operations are handled through standard user input devices (keyboard, mouse) and are executed on the application's host web server and the database server.

3.3 Software Interfaces

The system has one primary external software interface: the database.

- **Microsoft SQL Server Database**
  - **Description**: This is the existing, persistent data store for the live QTC WFQ system. It is the single source of truth for all workflow, step, and task data.
  - **Interface**: The WFQ Administrator Application will interface directly with the Workflow database using the provided schema. Data access will be managed via a data access layer within the application (e.g., using .NET Entity Framework) and by executing existing stored procedures where applicable (e.g., CreateWorkflowItem, GetNextTaskNumber). The application must be granted credentials with the necessary permissions to read from and write to the relevant tables.

3.4 Communications Interfaces

- **Protocol**: All communication between the administrator's web browser and the application's web server (IIS) shall be conducted securely over HTTPS/TLS.
- **Authentication:** User authentication is assumed to be handled by the hosting environment prior to the user accessing the application's URL. The application itself does not manage user logins or passwords. It only consumes the authorization status provided to it.

4. Requirements Specification

4.1 Functional Requirements

4.1.1 Authentication and Authorization

- 4.1.1.1 The system shall require all users to authenticate before accessing the Workflow Queue Administrator interface.
- 4.1.1.2 The system shall verify the user’s authorization level upon authentication.

- 4.1.1.3 The system shall restrict access to users with the “Admin” role.
- 4.1.1.4 The system shall deny access to users who do not have admin privileges, displaying an “Access Denied” message.
- 4.1.1.5 The system shall validate an admin authorization flag via the query string before granting access.

4.2 Workflow Management

4.2.1 Workflow Display

- 4.2.1.1 The system shall display a paginated list of all existing workflows.
- 4.2.1.2 Each workflow entry shall display: Name, Description, Status (New, In Progress, Completed), and Actions (Edit, View Steps).
- 4.2.1.3 The system shall allow admins to search or filter workflows by Name.
- 4.2.1.4 The system shall allow sorting by Name and CreatedDate.

4.2.2 Workflow Creation

- 4.2.2.1 The system shall allow an admin to create a new workflow.
- 4.2.2.2 The creation form shall include Name (required, unique) and Description (optional).
- 4.2.2.3 The system shall enforce a maximum length for the Name field.
- 4.2.2.4 The system shall validate that workflow names are unique.

4.2.3 Workflow Update

- 4.2.3.1 The system shall allow admins to update existing workflows.
- 4.2.3.2 The system shall prepopulate the update form with current workflow data.
- 4.2.3.3 The same validation rules as creation shall apply.
- 4.2.3.4 The system shall not allow workflow deletion through the interface.
- 4.2.3.5 The system shall display an informational note that deletion must be handled manually if required.

4.3 Workflow Step Management

4.3.1 Step Display

- 4.3.1.1 The system shall display all steps associated with a selected workflow.
- 4.3.1.2 Each step shall display: StepNumber, StepName, Description, and Actions (Edit/Delete).

4.3.2 Step Creation

- 4.3.2.1 The system shall allow the admin to create new steps within a workflow.
- 4.3.2.2 Required fields: StepName (unique within workflow).
- 4.3.2.3 Optional fields: StepNumber (auto-assigned if omitted).
- 4.3.2.4 The system shall validate StepNumber uniqueness within the workflow.

4.3.3 Step Update

- 4.3.3.1 The system shall allow admins to update existing step data.
- 4.3.3.2 The system shall prepopulate the step form with current information.
- 4.3.3.3 Validation rules shall match those used in creation.

4.3.4 Step Deletion

- 4.3.4.1 The system shall prompt for confirmation before deleting a step.
- 4.3.4.2 The system shall check for dependent tasks before deletion.
- 4.3.4.3 If dependencies exist, the system shall reassign tasks to the next step before deletion.
- 4.3.4.4 The system shall delete steps using Entity Framework, preventing cascading deletions beyond the step.

4.4 Task Management

- 4.4.1 The system shall display all tasks associated with a selected workflow.
- 4.4.2 Each task shall display: AssignedTo, Status, Step, and Actions (Complete Task / Reassign).
- 4.4.3 The system shall allow admins to assign or unassign tasks to users.
- 4.4.4 The system shall allow admins to mark tasks as complete.

4.5 Error Handling and Validation

- 4.5.1 The system shall perform input validation on both client and server sides.
- 4.5.2 The system shall display user-friendly error messages for invalid inputs.
- 4.5.3 The system shall handle and report database exceptions such as concurrency or foreign key violations.
- 4.5.4 Invalid or incomplete form submissions shall be rejected.

4.6 Workflow and Step History

- 4.6.1 The system shall display task and step history for each workflow.
- 4.6.2 History shall be shown in chronological order.
- 4.6.3 The system shall allow viewing history at both workflow and step levels in a readable format.

4.7 External Interface Requirements

Input Interfaces:

- **Location Search Input:** Allows users to enter service or location keywords (up to 100 characters).

Output Interfaces:

- **Map Display:** Displays nearby services using Google Maps API (interactive map with pins).

External APIs:

- Google Maps API for geolocation and rendering.
- Translation API for multilingual support.

4.8 Logical Database Requirements

- The system uses SQL Server and Oracle databases for managing workflows, steps, tasks, and users.
- Referential integrity and consistency must be maintained across both environments.

4.8.1 Core Entities

- **User:** UserID, Username, Email, Role, Status
- **Workflow:** WorkflowID, Name, Description, Status, CreatedBy, CreatedDate
- **WorkflowStep:** StepID, WorkflowID, StepNumber, StepName, Description
- **Task:** TaskID, WorkflowID, StepID, AssignedTo, Status, StartDate, CompletionDate

4.8.2 Constraints

- Each workflow must have at least one step.
- StepNumber must be unique within a workflow.
- Workflow names must be unique.
- Deleting a workflow must not delete dependent tasks.

4.8.3 Data Retention

- Completed workflows and task records must be retained for three years.
- Archived data will be read-only.

4.9 Design Constraints

- Built using ASP.NET MVC, C#, Entity Framework.
- Must run under IIS on Windows Server.
- Supports SQL Server 2019+ and Oracle 19c+.
- HTTPS/TLS 1.2+ required.
- Up to 100 concurrent users; page load ≤ 3 seconds.
- Developed in Visual Studio 2022+.
- Version control through GitHub Enterprise.
- Only approved NuGet packages permitted.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Average system response time should not exceed 2–3 seconds for any database query under normal conditions.
- The system shall handle concurrent requests from at least 100 users.
- Workflow search queries should return results within 1 second for up to 10,000 records.

5.2 Safety Requirements

- The system shall prevent unauthorized deletion or modification of workflow data.
- The application shall include automated daily backups of all critical data.
- In case of failure, the system shall restore from the last known good backup.

5.3 Security Requirements

- Authorization is based on a URL query string flag
- Role-based access control (RBAC) must be strictly enforced.
- Session timeout will occur after 15 minutes of inactivity.
- All sensitive fields (e.g., user credentials, role mappings) must be encrypted in transit and at rest.

5.4 Software Quality Attributes

- **Reliability:** The system must maintain 99.5% uptime during business hours.
- **Usability:** Interfaces must follow accessibility (Section 508) guidelines.
- **Maintainability:** Source code should follow SOLID design principles and include XML documentation.
- **Scalability:** The system must allow future integration with third-party APIs.
- **Portability:** The application must support deployment in both SQL Server and Oracle environments.

5.5 Business Rules

- Only administrators can create or modify workflows.
- Workflows cannot be deleted once created; they may only be archived.

- Tasks cannot be reassigned once marked as “Completed.”
- Workflow names must be unique and descriptive to prevent duplication.
- Each workflow must contain at least one step before activation.

6. Legal and Ethical Considerations

6.1 Scope and Sensitivity

The Workflow Queue (WFQ) system may process sensitive organizational data and personally identifiable information (PII) associated with authenticated users. The design therefore prioritizes confidentiality, integrity, availability, and accountability in both data handling and system behavior.

6.2 Access Control and Authorization

- All users authenticate through the organization's identity provider.
- Role-based access control (RBAC) enforces least-privilege access; only users with the Admin role may access administrative functions.
- Administrative actions are logged to support accountability and post-event review.

6.3 Transmission and Session Security

- All browser–server and service-to-service communications shall use HTTPS/TLS.
- Sessions shall follow organization policy for timeout and re-authentication (e.g., 15 minutes of inactivity).
- Sensitive values shall never be transmitted or stored in plaintext.
- Tokens, cookies, and session identifiers shall be marked HttpOnly and Secure and protected against CSRF.

6.4 Data Integrity, Auditability, and Non-Repudiation

- Workflow and task history shall be preserved and viewable to authorized users.
- Deleting workflows in the UI shall not be supported; archival is the approved lifecycle state.
- Database operations shall avoid cascade deletes that would remove dependent records.
- All administrative actions shall be logged with timestamp, actor, and outcome.

6.5 Retention and Archival

- Completed workflows and task records shall be retained per organizational retention policy and then archived in read-only form.
- Automated daily backups and documented restore procedures shall support continuity and evidence preservation.
- Access to archives shall be restricted and monitored.

6.6 Data Minimization and Purpose Limitation

- Only data required for defined business purposes shall be collected, stored, and exposed through interfaces.
- Service contracts shall limit attributes to the minimum necessary (e.g., identity and role).
- Write operations shall be constrained to approved pathways to reduce unintended data propagation.

6.7 Accessibility and Fairness

- User interfaces shall follow organizational accessibility standards (e.g., Section 508 awareness).
- Layouts, terminology, and error messaging shall be consistent and comprehensible to non-technical users.
- Features shall be designed to avoid bias in task routing, review, and assignment.

6.8 Compliance Alignment

- The system shall align with internal cybersecurity, authentication, and records-management policies.
- If scope later includes regulated data types (e.g., PHI), controls shall be extended to meet the applicable regulatory framework without weakening existing safeguards.
- Periodic reviews shall verify continued alignment with policy and regulatory changes.

6.9 Ethical Justification Summary

- The selected controls—least-privilege RBAC, TLS-only transport, secure sessions, immutable audit trails, retention plus read-only archival, and backups—collectively:
- Protect confidentiality and integrity.
- Uphold accountability and due process in task routing and approvals.
- Respect users through accessible, consistent interfaces.

Appendix A: Glossary

Term	Definition
WFQ	Workflow Queue System
RBAC	Role-Based Access Control
ORM	Object-Relational Mapping
HTTPS	Hypertext Transfer Protocol Secure
AD	Active Directory
MVC	Model-View-Controller
CRUD	Create, Read, Update, Delete

Appendix B: Analysis Models

- **Use Case Diagram:** Displays relationships between users (Admin, Manager, User) and system functions (Create Workflow, Approve Task, Track Progress).
- **ER Diagram:** Shows logical relationships between User, Workflow, WorkflowStep, and Task entities.
- **Sequence Diagram:** Illustrates user login and workflow approval sequence.

(Visual diagrams can be attached separately in Lucidchart or Visio format.)

Appendix C: To Be Determined List

Item	Description	Target Decision Date
Workflow Name Max Length	Maximum character limit for Name field	TBD by Dev Team
StepNumber Auto-Increment Rule	Define logic for assigning new step numbers	TBD by Design Lead
User Role Mapping Details	Final mapping between AD roles and WFQ roles	TBD by QTC Security Team
Reporting Module	Inclusion of dashboard analytics	Version 2.0
Notification Integration	Email and Teams alerts for approvals	Version 2.0